

Tunisian Darija NLP Reference Guide

For RAG-based chatbot intent detection and response adaptation

Version 1.0 — Built for Proxym demo

1. Arabizi Encoding System

Tunisian Darija is often written in Latin script using numbers to represent Arabic sounds that don't exist in French or English. This is called Arabizi. Two parallel encoding systems exist and both must be recognized.

Number	Letter alternative	Arabic letter	Sound	Example (Arabizi)	Example (Arabic)
9	q / k	ك	q (deep k)	9alet / qalet	كالايت
5	kh	خ	kh (guttural)	5oya / khoya	خويا
2	a / '	ا	glottal stop	2enti / aenti	ايتي
3	aa	آ	ayn (deep vowel)	3aych / aaych	ايش
7	h	ه	heavy h	7ajet / hajet	هايت
6	t	ط	emphatic t	6abi3i / tabi3i	طابي
8	gh	غ	gh (French r)	8ayeb / ghayeb	غاييب

Important: Both encoding variants must be treated as identical. '5oya' and 'khoya' mean the same word. Always normalize before matching.

2. Text Normalization Rules

Before any intent detection, apply these normalization steps in order:

Rule	Description	Example
Lowercase	Convert all text to lowercase	'3AWEDLY' → '3awedly'
Remove spaces	Strip all spaces (lazy typing skips them)	'ma fhemtch' → 'mafhemtch'
Strip repetition	Collapse 3+ repeated letters to 2	'haaaaannnn' → 'haann'
Fuzzy match	Use rapidfuzz with threshold ~80 for misspellings	'mrgl' matches 'mrgl'
Number→letter	Optionally expand numbers to letter equivalents	'5oya' → 'khoya'

3. Confusion / Did Not Understand

Trigger: model should ask for confirmation and offer to reformulate.

Darija (Arabizi)	Variants	Meaning	Aggression
mafhemt chy	mafhamtch / mafhemtch	I didn't understand anything	neutral
chnoa ?	chnou / shnou / chni	what?	neutral
zid fahamni	zid fehamni / zid fhamni	explain more	polite
zid fassarly	zid fesarly / zid fsarly	break it down more	polite
naarech	na3rech / naارش	huh? / what's that?	confused
ya5i fech te7ki	ya khi fech te7ki / ya5i shnou	bro what are you saying?	frustrated

bjaah rabby fech te7ki	b ja7 rabbi fech te7ki	for the love of god what?	frustrated
bjeh rabby fech t5arif	b ja7 rabbi fech tkharif	what are you saying (angry)	aggressive
yaa wildii	ya wlidi / ya weldi / ya wildi	my son (frustration signal)	frustrated

4. Repeat / Say Again Requests

Trigger: model should repeat its previous answer, in full or in part.

Darija (Arabizi)	Variants	Scope
3awidly	awidly / awedly / 3awedly	repeat anything
3awed miloul	3awed meloul / aawed miloul	repeat from the start
lfaza le5ra	lfaza lakhira / lfaza le5ra	the last part only
3awidly lfaza hethi	3awidly hathi / 3awedly hethi	repeat this specific part

5. Understood / Got It Signals

Trigger: model should proceed normally. Do NOT reformulate.

Darija (Arabizi)	Variants	Meaning	Note
haaannnnnn	hannn / hann / han	aha! got it	more letters = stronger confirmation
fhemtk	fhamtk / fhimtk / fhemtek	I understood you	
nfhem fik	nefhem fik / nifhem fik / nfhemfik	get you	space often dropped
mrgl	mrigl / mregl / mrigel	understood / noted	vowels often dropped
jawek behi	jawk bhi / jaw behi / jawk behi	your answer was good	
sa7it	sa7et / sa7 / sah	correct / nailed it	

Disambiguation note: 'mrgl' appears in both understood AND encouragement contexts. Check the preceding model turn — if it just answered a question, mrgl = understood. If it gave a long explanation, mrgl = encouragement.

6. Partially Understood

Trigger: model should ask what part was unclear and offer a simpler explanation.

Darija (Arabizi)	Variants	Meaning
fhemt ama mouch barcha	fhamt ama moch barcha / fhimt ama mch barcha	understood but not fully
mafhemtch mli7	mafhamt mli7 / mafhamtch mli7 / mafhemtch mli7	didn't understand well

7. Hallucination / Off Topic Signals

Trigger: model should acknowledge it went off topic and retry with the original question.

Darija (Arabizi)	Variants	Meaning
5arreeff	khareef / 5arif / kharif	you're hallucinating / off topic
mouch hakka	mouch haka / heka mouch kk / mch hakka	that's not it / you're wrong
mouch kk	moch kk / mch kk	it's not like that

Context rule: 'mouch hakka' after a model response = wrong answer. 'mouch kk' mid-conversation = disagreement. Check position in dialogue.

8. Simplify / Give Example Requests

Trigger: model should re-explain using a concrete example or simpler language.

Darija (Arabizi)	Variants	Meaning
a3tini example	a3tini mithal / a3tini mithel	give me an example
het example	hett mithal / het mithel / hett example	put an example
fassarly bisyassa	fesarly bisyassa / fsarly bisyassa	explain slowly / take it easy

9. Partial Agreement / Pushback

Trigger: model should acknowledge the pushback and address the specific disagreement.

Darija (Arabizi)	Variants	Meaning
ama	amma	yes but... (partial agreement)
mouch kk	moch kk / mch kk / mouch haka	it's not like that (correction)

10. Impatience / Too Long Signals

Trigger: model should give a shorter, more direct answer.

Darija (Arabizi)	Variants	Meaning
na9is mil7dith	na9es mel7dith / na9es mel hdith	less talking / get to the point

11. Giving Up / Abandoning the Question

Trigger: model should acknowledge, offer one last simplified attempt, then stop.

Darija (Arabizi)	Variants	Meaning
ouh 3lik ouh	ou3lik / ou 3lik	forget it / you're hopeless
la3alayna ayy	la3lina ay / la3leyna ayy	whatever / I give up
ayast moi	ayest / iyest / ayast	I'm exhausted / done trying

12. Swearing Detection Policy

Tunisian users swear frequently as emphasis or frustration markers — not literal insults. The chatbot must NOT be encouraged to swear or respond to swear words as meaningful intent signals.

Signal type	Action	Duration
Swear word detected	Pause response, show polite warning	10 second cooldown
Swear + confusion signal	Apply timeout THEN reformulate	10s then respond
Swear + giving up signal	Apply timeout THEN acknowledge	10s then respond

Implementation note: maintain a private SWEAR_WORDS list in your codebase. Strip swear words from the sentence before intent detection — treat them as punctuation, not meaning. Never log or repeat swear words back to the user.

13. Intent Detection Priority Order

When multiple signals appear in one message, process in this order:

Priority	Signal type	Action
1 (highest)	Swear word	Timeout (10s), then continue
2	Giving up	Acknowledge + one last attempt
3	Hallucination flag	Apologize + retry with original question
4	Off topic / wrong	Acknowledge + correct course
5	Confusion signal	Ask to confirm + offer reformulation
6	Half understood	Ask what part was unclear
7	Simplify request	Re-explain with example
8	Repeat request	Repeat full or partial answer
9	Impatience	Shorter answer
10 (lowest)	Understood / praise	Proceed normally

14. Sample Dialogues for RAG Context

These examples show how the chatbot should behave given specific Darija inputs. Include these in your RAG chunks so the model learns the expected response pattern.

User: mafhemt chy

Bot: ■■■ ■■■■■■■■ ■■■ — ■■■■■■ ■■■■ ■■■■■■■■ ■■■■■. [Reformulate in simpler terms]

User: zid fassarly bisyassa

Bot: ■■■■ — ■■■■ ■■■■. [Slow down, use shorter sentences]

User: 5arreeff

Bot: ■■■■ ■■■ ■■■■ ■■ ■■■■■■■■. ■■■■ ■■■■■■■■ ■■■■■■■■. [Reset to original question]

User: a3tini example

Bot: ■■■■■■ — ■■■ ■■■■ ■■■■: [Provide concrete example]

User: haaannnnnn fhemtk

Bot: [Proceed normally — user understood]

User: ama mouch kk

Bot: ■■■■■■ ■■■ ■■ ■■ ■■■■ ■■■■. [Invite correction]

User: ayast moi

Bot: ■■■■■■ — ■■■■■■ ■■■ ■■■■■■ ■■■■■ ■■■■■ ■■■■■. [One final simple attempt]

User: na9is mil7dith

Bot: [Give a 1-2 sentence direct answer only]

User: 3awed miloul

Bot: [Repeat full answer from the beginning]

User: lfaza le5ra 3awidha

Bot: [Repeat only the last paragraph/sentence]

This document is a RAG reference guide — chunk it by section for best retrieval performance. Each section should be a separate chunk so similarity search returns focused results.